

Convex Optimization 作业 1

1 For what value of α is the following matrix becomes positive definite?

$$A = \begin{vmatrix} 1 & \alpha & -1 \\ \alpha & 4 & 2 \\ -1 & 2 & 4 \end{vmatrix}$$

要确定矩阵 A 为正定矩阵，需要满足矩阵 A 的所有主子式 (leading principal minors) 都必须正的。

第一个主子式是 $A_{11} = 1$ ，显然是正的。

第二个主子式是左上角 2×2 子矩阵的行列式：

$$A_{22} = \begin{vmatrix} 1 & \alpha \\ \alpha & 4 \end{vmatrix} = 1 \cdot 4 - \alpha \cdot \alpha = 4 - \alpha^2$$

要使 A_{22} 为正，需要 $4 - \alpha^2 > 0$ ，即 $\alpha^2 < 4$ ，所以 $-2 < \alpha < 2$ 。

第三个主子式是整个矩阵 A 的行列式：

$$\det(A) = \begin{vmatrix} 1 & \alpha & -1 \\ \alpha & 4 & 2 \\ -1 & 2 & 4 \end{vmatrix}$$

计算这个行列式：

$$\begin{aligned} \det(A) &= 1 \cdot (4 \cdot 4 - 2 \cdot 2) - \alpha \cdot (\alpha \cdot 4 - 2 \cdot (-1)) - 1 \cdot (\alpha \cdot 2 - 4 \cdot (-1)) \\ &= 1 \cdot (16 - 4) - \alpha \cdot (4\alpha + 2) - 1 \cdot (2\alpha + 4) \end{aligned}$$

$$= 12 - 4\alpha^2 - 2\alpha - 2\alpha - 4$$

$$= 8 - 4\alpha^2 - 4\alpha$$

要使 $\det(A)$ 为正, 需要 $8 - 4\alpha^2 - 4\alpha > 0$, 即 $\alpha^2 + \alpha - 2 < 0$ 。

解这个不等式:

$$\alpha^2 + \alpha - 2 = 0$$

使用求根公式:

$$\alpha = \frac{-1 \pm \sqrt{1^2 - 4 \cdot 1 \cdot (-2)}}{2 \cdot 1}$$

$$\alpha = \frac{-1 \pm \sqrt{1+8}}{2}$$

$$\alpha = \frac{-1 \pm 3}{2}$$

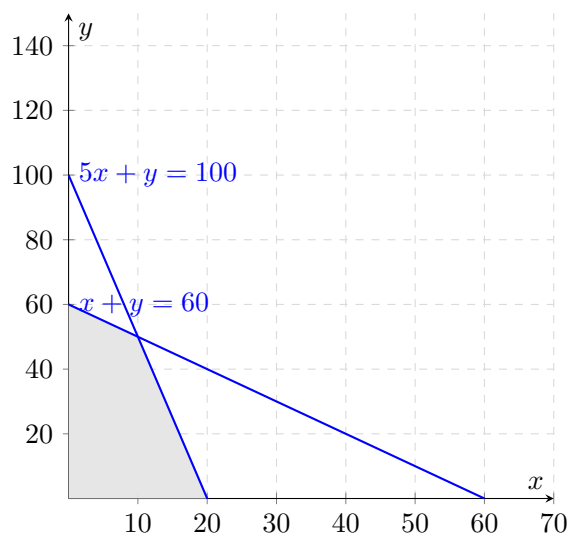
得到两个根 $\alpha = 1$ 和 $\alpha = -2$ 。因此, 不等式 $\alpha^2 + \alpha - 2 < 0$ 在 $-2 < \alpha < 1$ 时成立。

综合以上条件, α 的取值范围是 $-2 < \alpha < 1$ 。在这个范围内, 矩阵 A 是正定的。

2 Solve the following problem graphically

2.1 Maximize $f = 250x + 75y$, under the constraint

$$\begin{cases} 5x + y \leq 100 \\ x + y \leq 60 \\ x, y \geq 0 \end{cases}$$

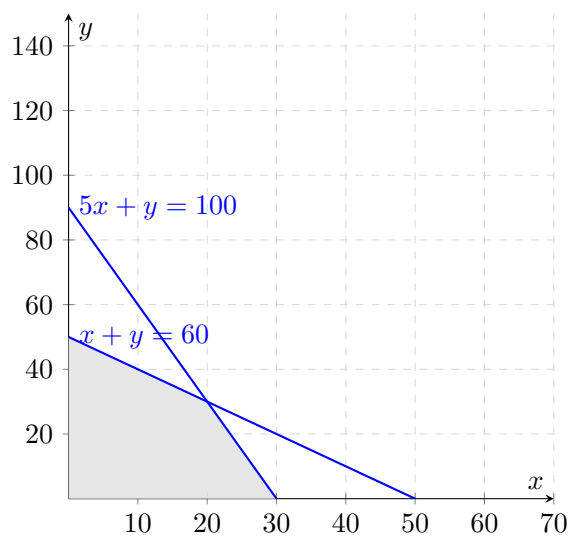


f 的斜率为-3.333, 介于-1 和-5 之间

经过平移 $f = 250x + 75y$ 可得当 $x=10, y=50$ 时 f 最大, $f=6250$

2.2 Maximize $f = 4x + y$, under the constraint

$$\begin{cases} x + y \leq 50 \\ 3x + y \leq 90 \\ x, y \geq 0 \end{cases}$$



$f=4x + y$ 斜率为-4, 大于-3, 所以当 $x=30, y=0$ 时 f 最大, $f=120$

- 3 A dietician wishes to mix two types of foods in such a way that vitamin contents of the mixture contain at least 8 units of vitamin A and 10 units of vitamin C. Food-I contains 2 units/kg of vitamin A and 1 unit/kg of vitamin C. Food-II contains 1 unit/kg of vitamin A and 2 units/kg of vitamin C. It costs 5/kg to purchase Food-I and 7/kg to purchase Food-II. Formulate this problem as a linear programming problem to minimize the cost of such a mixture.

线性规划问题:

$$\text{Minimize } Z = 5x_1 + 7x_2$$

subject to

$$\begin{cases} 2x_1 + x_2 \geq 8 \\ x_1 + 2x_2 \geq 10 \\ x_1 \geq 0 \\ x_2 \geq 0 \end{cases}$$

- We introduce two slack variables for two inequalities s_1, s_2

$$\text{Min. } 5x_1 + 7x_2$$

sub. to

$$\begin{cases} 2x_1 + x_2 + s_1 = 8 \\ x_1 + 2x_2 + s_2 = 10 \\ x_1, x_2 \geq 0 \\ s_1, s_2 \leq 0 \end{cases}$$

- For the 1st constraint, $2x_1$ should be no greater than 8, $x_1 \leq 4$
- For the 2nd constraint, x_1 should be no greater than 10, $x_1 \leq 10$
- We can see x_1 should be increased to 4 at most
- With the 1st constraint, we have $x_1 = 4 - 0.5 * s_1 - 0.5 * x_2$

$$\text{Min. } 20 + 4.5x_2 - 2.5s_1$$

sub. to

$$\begin{cases} 2x_1 + x_2 + s_1 = 8 \\ 1.5x_2 - 0.5s_1 + s_2 = 6 \\ x_1, x_2 \geq 0 \\ s_1, s_2 \leq 0 \end{cases}$$

- For the 1st constraint, x_2 should be no greater than 8, $x_2 \leq 8$
- For the 2nd constraint, x_2 should be no greater than 4, $x_2 \leq 4$
- We can see x_2 should be increased to 4 at most
- With the 2nd constraint, we have $x_2 = 4 + \frac{1}{3} * s_1 - \frac{2}{3} * s_2$

$$\text{Min. } 38 - s_1 - 3s_2$$

sub. to

$$\begin{cases} 2x_1 + \frac{4}{3}s_1 - \frac{2}{3}s_2 = 4 \\ 1.5x_2 - 0.5s_1 + s_2 = 6 \\ x_1, x_2 \geq 0 \\ s_1, s_2 \leq 0 \end{cases}$$

- Plug $s_1 = 0$ and $s_2 = 0$ into the first two constraints
- We have $x_1 = 2, x_2 = 4, \text{answer} = 38$

- 4 There are two factories located one at place P and the other at place Q. From these locations, a certain commodity is to be delivered to each of the three depots situated at A, B and C. The weekly requirements of the depots are respectively 5, 5 and 4 units of the commodity while the production capacity of the factories at P and Q are respectively 8 and 6 units. The cost of transportation per unit is given in Table 1. Formulate this problem as a linear programming problem to minimize the cost of transportation.

表 1: Transportation Costs in Dollars

From/To	A	B	C
P	16	10	15
Q	10	12	10

线性规划问题：最小化运输成本

- 定义决策变量：设 x_{ij} 为从工厂 i 到仓库 j 的运输量，其中 $i = P, Q$, $j = A, B, C$ 。
- 目标函数：最小化总运输成本

$$Z = \sum_{i,j} c_{ij} x_{ij}$$

其中 c_{ij} 是从工厂 i 到仓库 j 的单位运输成本。

目标函数：

$$\text{Minimize } Z = 16x_{PA} + 10x_{PB} + 15x_{PC} + 10x_{QA} + 12x_{QB} + 10x_{QC}$$

约束条件：

$$\left\{ \begin{array}{l} x_{PA} + x_{PB} + x_{PC} \leq 8 \\ x_{QA} + x_{QB} + x_{QC} \leq 6 \\ x_{PA} + x_{QA} \geq 5 \\ x_{PB} + x_{QB} \geq 5 \\ x_{PC} + x_{QC} \geq 4 \\ x_{PA}, x_{PB}, x_{PC}, x_{QA}, x_{QB}, x_{QC} \geq 0 \end{array} \right.$$